13.2.2 Modular programming:-

Modular programming consist of a list of instructions that instructs the computer to do something. But this **Paradigm consists of multiple modules, each module has a set of functions of related types. Data is hidden under the modules.** Arrangement of data can be changed only by modifying the module

Important features of Modular programming

- Emphasis on algorithm rather than data
- Programs are divided into individual modules
- Each modules are independent of each other and have their own local data
- Modules can work with its own data as well as with the data passed to it.
- Example: **Pascal** and **C**

13.2.3 Object Oriented Programming:-

Object Oriented Programming paradigm emphasizes on the data rather than the algorithm. It implements programs using **classes** and **objects.**

Class: A Class is a construct in C++ which is used to bind data and its associated function together into a single unit using the encapsulation concept. Class is a user defined data type. Class represents a group of similar objects.

It can also be defined as a template or blueprint representing a group objects that share common properties and relationship.

Objects: Represents data and its associated function together into a single unit. Objects are the basic unit of OOP. Basically an object is created from a class. They are instances of class also called as class variables

An identifiable entity with some characteristics and behaviour is called object.

Important features of Object oriented programming

- Emphasizes on data rather than algorithm
- Data abstraction is introduced in addition to procedural abstraction
- Data and its associated operations are grouped in to single unit
- Programs are designed around the data being operated
- Relationships can be created between similar, yet distinct data types
- Example: **C++, Java, VB.Net, Python etc.**

13.3 Basic Concepts of OOP

The Object Oriented Programing has been developed to overcome the drawbacks of procedural and modular programming. It is widely accepted that object-oriented programming is the most important and powerful way of creating software.

The Object-Oriented Programming approach mainly encourages:

- **Modularisation:** where the program can be decomposed into **modules**.
- **Software re-use:** where a program can be composed from existing and new modules.

Main Features of Object Oriented Programming

- Data Abstraction
- Encapsulation

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